# MATHEMATICS

# *TRIGONOMETRIC RATIO*
# *THEORY SHEET*
# *ENGLISH*

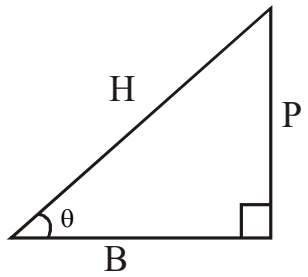
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**MATRIX**

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## 1.1 INTRODUCTION TO TRIGONOMETRIC RATIOS

In the right angled triangle OMP , we have base (OM) = x, perpendicular (PM) = y and hypotenuse (OP) = r, then we define the following trigonometric ratios which are known as trigonometric function.




$\sin\theta = \frac{P}{H} = \frac{y}{r}$ = sine of $\theta$

$\cos\theta = \frac{B}{H} = \frac{x}{r}$ = cosine of $\theta$

$\tan\theta = \frac{P}{B} = \frac{y}{x}$ = tangent of $\theta$

$\cot\theta = \frac{B}{P} = \frac{x}{y}$ = cotangent of $\theta$

$\sec\theta = \frac{H}{B} = \frac{r}{x}$ = secant of $\theta$

$\operatorname{cosec}\theta = \frac{H}{P} = \frac{r}{y}$ = cosecant of $\theta$

**Fundamental Trigonometrical Identities :**

1. $\sin^2\theta + \cos^2\theta = 1$
2. $\sec^2\theta - \tan^2\theta = 1$
3. $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

**Questions**

1. Prove that $\sin^4\theta + \cos^4\theta = 1 - 2\cos^2\theta \sin^2\theta$.

Sol. $\sin^4\theta + \cos^4\theta = (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta \cos^2\theta$

$= 1 - 2\sin^2\theta \cos^2\theta$

2. Prove that $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta \cos^2\theta$.

Sol. $\sin^6\theta + \cos^6\theta = (\sin^2\theta + \cos^2\theta)(\sin^4\theta + \cos^4\theta - \sin^2\theta \cos^2\theta)$

$= (1)(1 - 2\sin^2\theta \cos^2\theta - \sin^2\theta \cos^2\theta)$

$= 1 - 3\sin^2\theta \cos^2\theta$

3. Prove that $\frac{\tan^2\theta . \sin^2\theta}{\tan^2\theta - \sin^2\theta} = 1$

Sol. $$\frac{\tan^2\theta\cdot\sin^2\theta}{\tan^2\theta-\sin^2\theta}\Rightarrow\frac{\frac{\sin^2\theta}{\cos^2\theta}\cdot\sin^2\theta}{\frac{\sin^2\theta}{\cos^2\theta}-\sin^2\theta}$$

$$=\frac{\frac{\sin^4\theta}{\cos^2\theta}}{\sin^2\theta\left(\frac{1}{\cos^2\theta}-1\right)}\Rightarrow\frac{\frac{\sin^4\theta}{\cos^2\theta}}{\sin^2\theta\left(\frac{1-\cos^2\theta}{\cos^2\theta}\right)}$$

$$=\frac{\frac{\sin^4\theta}{\cos^2\theta}}{\frac{\sin^2\theta\cdot\sin^2\theta}{\cos^2\theta}}=1$$

4. Prove that $\frac{\cot^2\theta.\cos^2\theta}{\cot^2\theta-\cos^2\theta}=1$

Sol. $$\frac{\cot^2\theta\cdot\cos^2\theta}{\cot^2\theta-\cos^2\theta}\Rightarrow\frac{\frac{\cos^2\theta}{\sin^2\theta}\cdot\cos^2\theta}{\frac{\cos^2\theta}{\sin^2\theta}-\cos^2\theta}$$

$$=\frac{\frac{\cos^4\theta}{\sin^2\theta}}{\cos^2\theta\left(\frac{1-\sin^2\theta}{\sin^2\theta}\right)}\Rightarrow\frac{\frac{\cos^4\theta}{\sin^2\theta}}{\frac{\cos^2\theta.\cos^2\theta}{\sin^2\theta}}\Rightarrow1$$

5. $\sec\theta+\tan\theta=2$. find $\sin\theta$

Ans. $\frac{3}{5}$

Sol. $\sec\theta+\tan\theta=2$

$\sec\theta-\tan\theta=\frac{1}{2}$

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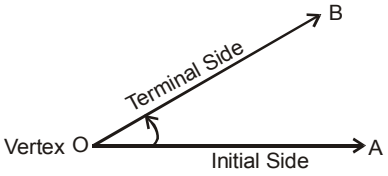
$2\sec\theta=\frac{5}{2}$

$\sec\theta=\frac{5}{4}$

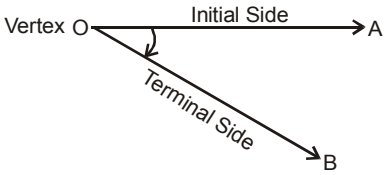
$\cos\theta=\frac{4}{5}$

$\sin\theta=\frac{3}{5}$

## 1.2 MEASUREMENT OF ANGLE




(i) Positive angle
(anticlockwise measurement)



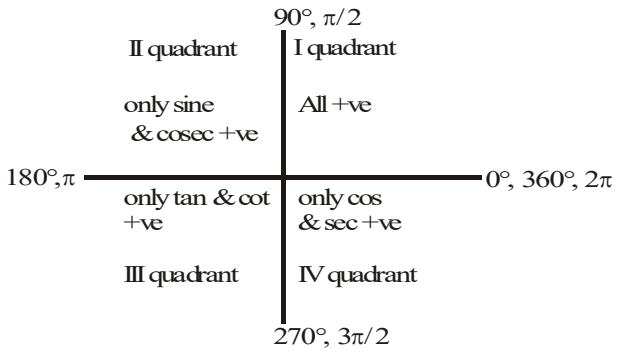

(ii) Negative angle
(clockwise measurement)

One right angle = 90°, $1° = 60'$, $1' = 60''$

$\pi$ radian = 180°

**Note :**

If no symbol is mentioned while showing measurement of angle, then it is considered to be measured in radians.




## 2.1 REDUCTION FORMULAE

Imp. angle + $\theta$

Imp. angle lies on horizontal axis

Imp. angle lies on verticle axis

$\sin(\pi+\theta) = -\sin\theta$ $\qquad$ $\sin\left(\frac{\pi}{2}+\theta\right) = +\cos\theta$

$\cos(\pi-\theta) = -\cos\theta$ $\qquad$ $\cos\left(\frac{3\pi}{2}-\theta\right) = -\sin\theta$

$\tan(\pi+\theta) = +\tan\theta$ $\qquad$ $\tan\left(\frac{3\pi}{2}+\theta\right) = -\cot\theta$

$\cot(2\pi-\theta) = -\cot\theta$ $\qquad$ $\cos\left(\frac{3\pi}{2}+\theta\right) = +\sin\theta$

- **Trigonometric Ratios of $\left(\frac{\pi}{2}-\theta\right)$:**

$\sin\left(\frac{\pi}{2}-\theta\right) = \cos\theta$, $\qquad$ $\tan\left(\frac{\pi}{2}-\theta\right) = \cot\theta$, $\qquad$ $\text{cosec}\left(\frac{\pi}{2}-\theta\right) = \sec\theta$,

$\cos\left(\frac{\pi}{2}-\theta\right) = \sin\theta$, $\qquad$ $\cot\left(\frac{\pi}{2}-\theta\right) = \tan\theta$, $\qquad$ $\sec\left(\frac{\pi}{2}-\theta\right) = \text{cosec}\,\theta$

- **Trigonometric Ratios of $\left(\frac{\pi}{2}+\theta\right)$**

$\sin\left(\frac{\pi}{2}+\theta\right)=\cos\theta,$ $\tan\left(\frac{\pi}{2}+\theta\right)=-\cot\theta,$ $\operatorname{cosec}\left(\frac{\pi}{2}+\theta\right)=\sec\theta,$

$\cos\left(\frac{\pi}{2}+\theta\right)=-\sin\theta,$ $\cot\left(\frac{\pi}{2}+\theta\right)=-\tan\theta,$ $\sec\left(\frac{\pi}{2}+\theta\right)=-\operatorname{cosec}\theta$

- **Trigonometric Ratios of $\pi-\theta$**

$\sin(\pi-\theta)=\sin\theta$ $\tan(\pi-\theta)=-\tan\theta$ $\sec(\pi-\theta)=-\sec\theta$

$\cos(\pi-\theta)=-\cos\theta$ $\cot(\pi-\theta)=-\cot\theta$ $\operatorname{cosec}(\pi-\theta)=\operatorname{cosec}\theta$

- **Trigonometric Ratios of $(\pi+\theta)$**

$\sin(\pi+\theta)=-\sin\theta,$ $\tan(\pi+\theta)=\tan\theta,$ $\operatorname{cosec}(\pi+\theta)=-\operatorname{cosec}\theta,$

$\cos(\pi+\theta)=-\cos\theta,$ $\cot(\pi+\theta)=\cot\theta,$ $\sec(\pi+\theta)=-\sec\theta$

- **Trigonometric Ratios of $\left(\frac{3\pi}{2}-\theta\right)$**

$\sin\left(\frac{3\pi}{2}-\theta\right)=-\cos\theta,$ $\tan\left(\frac{3\pi}{2}-\theta\right)=\cot\theta,$ $\operatorname{cosec}\left(\frac{3\pi}{2}-\theta\right)=-\sec\theta$

$\cos\left(\frac{3\pi}{2}-\theta\right)=-\sin\theta,$ $\cot\left(\frac{3\pi}{2}-\theta\right)=\tan\theta,$ $\sec\left(\frac{3\pi}{2}-\theta\right)=-\operatorname{cosec}\theta,$

- **Trigonometric Ratios of $\left(\frac{3\pi}{2}+\theta\right)$**

$\sin\left(\frac{3\pi}{2}+\theta\right)=-\cos\theta,$ $\cos\left(\frac{3\pi}{2}+\theta\right)=\sin\theta,$ $\tan\left(\frac{3\pi}{2}+\theta\right)=-\cot\theta,$

$\cot\left(\frac{3\pi}{2}+\theta\right)=-\tan\theta,$ $\sec\left(\frac{3\pi}{2}+\theta\right)=\operatorname{cosec}\theta,$ $\operatorname{cosec}\left(\frac{3\pi}{2}+\theta\right)=-\sec\theta$

- **Trigonometric Ratios of $(2\pi-\theta)$**

$\sin(2\pi-\theta)=-\sin\theta$ $\tan(2\pi-\theta)=-\tan\theta$ $\sec(2\pi-\theta)=\sec\theta$

$\cos(2\pi-\theta)=\cos\theta$ $\cot(2\pi-\theta)=-\cot\theta$ $\operatorname{cosec}(2\pi-\theta)=\operatorname{cosec}\theta$

- **Trigonometric Ratios of $(2\pi+\theta)$**

$\sin(2\pi+\theta)=\sin\theta$ $\tan(2\pi+\theta)=\tan\theta$ $\sec(2\pi+\theta)=\sec\theta$

$\cos(2\pi+\theta)=\cos\theta$ $\cot(2\pi+\theta)=\cot\theta$ $\operatorname{cosec}(2\pi+\theta)=\operatorname{cosec}\theta$

- **Trigonometric Ratios of $(-\theta)$:**

$\sin(-\theta)=-\sin\theta.$ $\tan(-\theta)=-\tan\theta.$ $\sec(-\theta)=\sec\theta.$

$\cos(-\theta)=\cos\theta.$ $\cot(-\theta)=-\cot\theta.$ $\operatorname{cosec}(-\theta)=-\operatorname{cosec}\theta.$

| | 0 | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\frac{2\pi}{3}$ | $\frac{5\pi}{6}$ | $\pi$ | $\frac{7\pi}{6}$ | $\frac{4\pi}{3}$ | $\frac{3\pi}{2}$ | $\frac{5\pi}{3}$ | $\frac{11\pi}{6}$ | $2\pi$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| sin | 0 | $\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1 | | | | | | | | | |
| cos | 1 | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$ | 0 | | | | | | | | | |
| tan | 0 | $\frac{1}{\sqrt{3}}$ | 1 | $\sqrt{3}$ | N.D. | | | | | | | | | |

## 2.2 IMPORTANT POINTS

1. $\sin(n\pi) = 0$ $\quad \sin\left(2n\pi + \frac{\pi}{2}\right) = 1$; $\quad \sin\left(2n\pi - \frac{\pi}{2}\right) = -1$

$\cos\left((2n+1)\frac{\pi}{2}\right) = 0$ $\quad \cos(2n\pi) = 1$ $\quad \cos((2n+1)\pi) = -1$

2. Complementary & Supplementary observation

If A & B are complementry angles $\quad A + B = \pi/2$

$\sin A = \sin\left(\frac{\pi}{2} - B\right) = \cos B$; $\quad \cos A = \sin B$; $\quad \tan A = \cot B$ ; $\quad \cot A = \tan B$

If A & B are supplementry angles $\quad A + B = \pi$

$\sin A = \sin(\pi - B) = \sin B$ $\quad \cot A = -\cot B$

$\cos A = \cos(\pi - B) = -\cos B$ $\quad \sec A = -\sec B$

$\tan A = -\tan B$ $\quad \operatorname{cosec} A = \operatorname{cosec} B$

Q.1 $\sin^2\left(\frac{\pi}{8}\right) + \sin^2\left(\frac{3\pi}{8}\right) + \sin^2\left(\frac{5\pi}{8}\right) + \sin^2\left(\frac{7\pi}{8}\right)$

Ans. 2

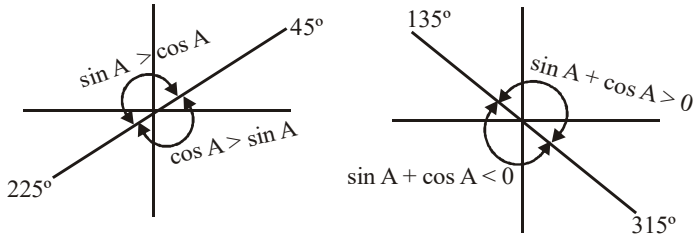
Sol. $\sin^2\left(\frac{\pi}{8}\right) + \sin^2\left(\frac{\pi}{2} - \frac{\pi}{8}\right) + \sin^2\left(\frac{\pi}{2} + \frac{\pi}{8}\right) + \sin^2\left(\pi - \frac{\pi}{8}\right)$

$\sin^2\frac{\pi}{8} + \cos^2\frac{\pi}{8} + \cos^2\frac{\pi}{8} + \sin^2\frac{\pi}{8}$

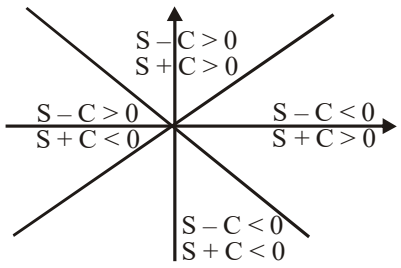
$= 1 + 1$

$= 2$

3.




Eg:– $|\sin 204^\circ - \cos 204^\circ| = \sin 204^\circ - \cos 204^\circ$



## 3.1 COMPOUND ANGLES

1. $\sin (A + B) = \sin A \cos B + \cos A \sin B$
2. $\sin(A - B) = \sin A \cos B - \cos A \sin B$
3. $\cos (A + B) = \cos A \cos B - \sin A \sin B$
4. $\cos (A - B) = \cos A \cos B + \sin A \sin B$

Adding (1) and (2),

5. $2 \sin A \cos B = \sin (A + B) + \sin (A - B)$

Subtracting (2) from (1),

6. $2 \cos A \sin B = \sin (A + B) - \sin (A - B)$

Adding (3) and (4),

7. $2 \cos A \cos B = \cos (A + B) + \cos (A - B)$

Subtraction (3) from (4).

8. $2 \sin A \sin B = \cos (A - B) - \cos (A + B)$

Let $A + B = C$ and $A - B = D$ then $A = \frac{C+D}{2}$ and $B = \frac{C-D}{2}$

Substituting in (5),

9. $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right).\cos\left(\frac{C-D}{2}\right)$

similarly other formula are,

10. $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right).\sin\left(\frac{C-D}{2}\right)$

11. $\cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right).\cos\left(\frac{C-D}{2}\right)$

12. $\cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right).\sin\left(\frac{D-C}{2}\right)$

13. $\sin (A + B).\sin (A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
14. $\cos (A + B).\cos (A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

**Questions**

1. Prove that $\frac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \tan \theta$.

Sol. $\frac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} \Rightarrow \frac{2\cos\left(\frac{7\theta+5\theta}{2}\right)\sin\left(\frac{7\theta-5\theta}{2}\right)}{2\cos\left(\frac{7\theta+5\theta}{2}\right)\cos\left(\frac{7\theta-5\theta}{2}\right)}$

$= \frac{\sin \theta}{\cos \theta} = \tan \theta$

2. Find value of $\sin^2(105^\circ) - \sin^2(15^\circ)$

Ans. $\frac{\sqrt{3}}{2}$

Sol. $\sin^2(105) - \sin^2(15^\circ)$

$\sin^2(90+15) - \sin^2 15^\circ$ $\qquad$ $\cos(15+15)\,.\,\cos(15\,.\,15)$

$\cos^2 15^\circ - \sin^2 15^\circ$ $\qquad$ $\cos 30^\circ.\cos 0^\circ$

$\cos^2(15^\circ)$ $\qquad$ $\frac{\sqrt{3}}{2}\cdot 1 \Rightarrow \frac{\sqrt{3}}{2}$

$\cos 30^\circ \Rightarrow \frac{\sqrt{3}}{2}$

3. Prove that $\frac{(\cos\theta - \cos 3\theta)(\sin 8\theta + \sin 2\theta)}{(\sin 5\theta - \sin\theta)(\cos 4\theta - \cos 6\theta)} = 1$

Sol. $\frac{(\cos\theta - \cos 3\theta)(\sin 8\theta + \sin 2\theta)}{(\sin 5\theta - \sin\theta)(\cos 4\theta - \cos 6\theta)}$

$$\frac{\left[2\sin\left(\frac{3\theta-\theta}{2}\right)\sin\left(\frac{\theta+3\theta}{2}\right)\right]\left[2\sin\left(\frac{8\theta+2\theta}{2}\right)\cos\left(\frac{8\theta-2\theta}{2}\right)\right]}{\left[2\cos\left(\frac{5\theta+\theta}{2}\right)\sin\left(\frac{5\theta-\theta}{2}\right)\right]\left[2\sin\left(\frac{6\theta-4\theta}{2}\right)\sin\left(\frac{6\theta+4\theta}{2}\right)\right]}$$

$$\frac{\sin\theta\sin 2\theta\sin 5\theta\cos 3\theta}{\cos 3\theta\sin 2\theta\sin\theta\sin 5\theta} = 1$$

4. Prove that $\frac{\sin A + 2\sin 3A + \sin 5A}{\sin 3A + 2\sin 5A + \sin 7A} = \frac{\sin 3A}{\sin 5A}$

Sol. $\frac{2\sin 3A\cos 2A + 2\sin 3A}{2\sin 5A\cos 2A + 2\sin 5A} \Rightarrow \frac{2\sin 3A(\cos 2A+1)}{2\sin 5A(\cos 2A+1)}$

$= \frac{\sin 3A}{\sin 5A}$

5. Prove that $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$

Sol. $\frac{2\sin 3A\cos 2A + 2\sin 5A\cos 2A}{2\cos 3A\cos 2A + 2\cos 5A\cos 2A}$

$\frac{2\cos 2A(\sin 3A + \sin 5A)}{2\cos 2A(\cos 3A + \cos 5A)}$

$= \frac{2\sin 4A\cos A}{2\cos 4A\cos A} \Rightarrow \tan 4A$

6. Prove that $\sin 10^\circ - \sin 70^\circ + \sin 50^\circ = 0$.

Sol. $\sin 10^\circ + \sin 50^\circ - \sin 70^\circ$

$2\sin 30^\circ \cos 20^\circ - \sin 70^\circ$

$= 2\times\frac{1}{2}\cos 20^\circ - \sin 70^\circ$

$= \sin 70^\circ - \sin 70^\circ$

$= 0$

15. $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ 16. $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

17. $\cot(A+B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$ 18. $\cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$

19. $\tan\left(\frac{\pi}{4} + A\right) = \frac{1+\tan A}{1-\tan A}$ 20. $\tan\left(\frac{\pi}{4} - A\right) = \frac{1-\tan A}{1+\tan A}$

## 3.2 2A FORMULAE

1. (i) $\sin 2A = 2\sin A \cos A$ (ii) $\sin 2A = \frac{2\tan A}{1+\tan^2 A}$

2. (i) $\cos 2A = \cos^2 A - \sin^2 A$ (ii) $\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$

$= 2\cos^2 A - 1$

$= 1 - 2\sin^2 A$

3. (i) $(\sin A + \cos A)^2 = 1 + \sin 2A$ (ii) $(\sin A - \cos A)^2 = 1 - \sin 2A$

(iii) $2\cos^2 A = 1 + \cos 2A$ (iv) $2\sin^2 A = 1 - \cos 2A$

4. $\tan 2A = \frac{2\tan A}{1-\tan^2 A}$

5. (i) $\tan^2 A = \frac{1-\cos 2A}{1+\cos 2A}$ (ii) $\tan A = \frac{1-\cos 2A}{\sin 2A}$ (iii) $\tan A = \frac{\sin 2A}{1+\cos 2A}$

6. (i) $\tan 15^\circ = 2 - \sqrt{3}$ (ii) $\tan 75^\circ = 2 + \sqrt{3}$

(iii) $\tan(22.5^\circ) = \sqrt{2} - 1$ (iv) $\tan(67.5^\circ) = \sqrt{2} + 1$

(v) $\sin 15^\circ$ or $\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$ or $\cos\frac{5\pi}{12}$

(vi) $\cos 15^\circ$ or $\cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ$ or $\sin\frac{5\pi}{12}$

**Questions**

1. $\tan\alpha = \frac{5}{6}$, $\alpha + \beta = \frac{\pi}{4}$ .Find $\tan\beta$.

Ans. $\frac{1}{11}$

Sol. $\tan(\alpha+\beta)=1$

$$\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}=1$$

$$\frac{5}{6}+\tan\beta=1-\frac{5}{6}\tan\beta$$

$$\tan\beta+\frac{5}{6}\tan\beta=1-\frac{5}{6}$$

$$\frac{11}{6}\tan\beta=\frac{1}{6}$$

$$\tan\beta=\frac{1}{11}$$

2. Prove that $\frac{1+\sin 2A}{\cos 2A}=\tan\left(\frac{\pi}{4}+A\right)$.

Sol. $\frac{1+\sin^2 A}{\cos 2A}\Rightarrow\frac{1+2\sin A\cos A}{\cos^2 A-\sin^2 A}$

$$=\frac{(\cos A+\sin A)^2}{(\cos A-\sin A)(\cos A+\sin)}$$

$$=\frac{\cos A+\sin A}{\cos A-\sin A}\div\frac{\cos A}{\cos A}$$

$$=\frac{1+\tan A}{1-\tan A}\Rightarrow\tan\left(\frac{\pi}{4}+A\right)$$

3. If $\cos A=\frac{1}{3}$ find $\cos 2A$.

Ans. $\frac{-7}{9}$

Sol. $\cos2A=2\cos^2A-1$

$$=2\left(\frac{1}{3}\right)^2-1\Rightarrow\frac{2}{9}-1$$

$$=\frac{-7}{9}$$

4. If $\sin A=\frac{2}{3}$ find $\cos 2A$.

Ans. $\frac{1}{9}$

Sol. $\cos2A=1-2\sin^2A$

$= 1 - 2\left(\frac{2}{3}\right)^2$

$= 1 - \frac{8}{9} \Rightarrow \frac{1}{9}$

5. If $\sin A = \frac{3}{5}$ find $\sin 2A$.

Ans. $\pm\frac{24}{25}$

Sol. $\cos A = \pm\frac{4}{5}$

$\sin 2A = 2\sin A \cos A$

$= 2 \times \frac{3}{5} \times \frac{4}{5} \Rightarrow \pm\frac{24}{25}$

6. If $\tan A = \frac{1}{3}$ find

(i) $\sin 2A$ (ii) $\cos 2A$ (iii) $\tan 2A$

Ans. (i) $\frac{3}{5}$ (ii) $\frac{4}{5}$ (iii) $\frac{3}{4}$

Sol. (i) $\sin^2 A = \frac{2\tan A}{1+\tan^2 A}$ $\Rightarrow \frac{2\left(\frac{1}{3}\right)}{1+\left(\frac{1}{3}\right)^2} \Rightarrow \frac{\frac{2}{3}}{\frac{10}{9}}$

$= \frac{3}{5}$

(ii) $\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$ $\Rightarrow \frac{1-\left(\frac{1}{3}\right)^2}{1+\left(\frac{1}{3}\right)^2} \Rightarrow \frac{\frac{8}{9}}{\frac{10}{9}}$

$= \frac{4}{5}$

(iii) $\tan 2A = \frac{2\tan A}{1-\tan^2 A} \Rightarrow \frac{2\left(\frac{1}{3}\right)}{1-\left(\frac{1}{3}\right)^2} \Rightarrow \frac{\frac{2}{3}}{\frac{8}{9}}$

$= \frac{3}{4}$

7. If $\cos A = \frac{7}{25}$ ; $A \in \left(0, \frac{\pi}{2}\right)$ find

(i) $\sin \frac{A}{2}$ (ii) $\cos \frac{A}{2}$ (iii) $\sin \frac{A}{4}$

Ans. (i) $\frac{3}{5}$ (ii) $\frac{4}{5}$ (iii) $\frac{1}{\sqrt{10}}$

Sol. (i) $\cos A = 1 - 2\sin^2 \frac{A}{2}$

$2\sin^2 \frac{A}{2} = 1 - \cos A$

$2\sin^2 A = 1 - \frac{7}{25} \Rightarrow \frac{18}{25}$

$\sin^2 A = \frac{9}{25}$

$\sin A = \frac{3}{5}$

(ii) $\cos A = 2\cos^2 \frac{A}{2} - 1$

$\frac{7}{25} + 1 = 2\cos^2 \frac{A}{2}$

$\frac{32}{25} = 2\cos^2 \frac{A}{2}$

$\frac{4}{5} = \cos \frac{A}{2}$

(iii) $\cos \frac{A}{2} = 1 - 2\sin^2 \frac{A}{4}$

$\frac{4}{5} - 1 = -2\sin^2 \frac{A}{4}$

$\frac{1}{10} = \sin^2 \frac{A}{4}$

$\frac{1}{\sqrt{10}} = \sin \frac{A}{4}$

8. $f(x) = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1} \forall\, x \in \left(0, \frac{\pi}{2}\right)$; find range of $f(x)$.

Ans. $(1, \infty)$

Sol. $f(x) = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1} \Rightarrow \frac{\sec x + \tan x - \left(\sec^2 x - \tan^2 x\right)}{\tan x - \sec x + 1}$

$$= \frac{(\sec x + \tan x) - [(\sec x + \tan x)(\sec x - \tan x)]}{\tan x - \sec x + 1}$$

$$= \frac{(\sec x + \tan x)(1 - \sec x + \tan x)}{\tan x - \sec x + 1}$$

$$= \sec x + \tan x \quad x \in \left(0, \frac{\pi}{2}\right)$$

$f(x) \in (1, \infty)$

9. Prove that $\frac{\sec(8A) - 1}{\sec(4A) - 1} = \frac{\tan 8A}{\tan 2A}$.

Sol. $$\frac{\sec 8A - 1}{\sec 4A - 1} \Rightarrow \frac{\frac{1 - \cos 8A}{\cos 8A}}{\frac{1 - \cos 4A}{\cos 4A}} \Rightarrow \frac{\frac{2\sin^2 4A}{\cos 8A}}{\frac{2\sin^2 2A}{\cos 4A}}$$

$$= \frac{2\sin 4A \cos 4A \sin 4A}{2\sin^2 2A \cos 8A} \Rightarrow \frac{\sin 8A \sin 4A}{2\sin^2 A \cos 8A \sin^2 A}$$

$$= \frac{\tan 8A 2\sin^2 A \cos^2 A}{2\sin^2 A \sin^2 A} \Rightarrow \frac{\tan 8A}{\tan 2A}$$

10. Prove that $(1 + \cos 2A + \sin 2A)^2 = 4\cos^2 A(1 + \sin 2A)$

Sol. $(1 + \cos 2A + \sin 2A)^2$

$(2\cos^2 A + 2\sin A \cos A)^2$

$= 4\cos^2 A (\cos A + \sin A)^2$

$= 4\cos^2 A (1 + \sin^2 A)$

11. Prove that $\sqrt{3}$ cosec 20º – sec 20º = 4.

Sol. $$\frac{\sqrt{3}}{\sin 20°} - \frac{1}{\cos 20°} \Rightarrow \frac{\sqrt{3}\cos 20 - \sin 20°}{\sin 20° \cos 20°}$$

$$= 2\frac{\left[\frac{\sqrt{3}}{2}\cos 20 - \frac{1}{2}\sin 20°\right]}{\sin 20° \cos 20°} \Rightarrow 2.2\left[\frac{\sin 60^\circ \cos 20^\circ - \cos 60^\circ \sin \frac{\pi}{2}}{2\sin 20^\circ \cos 20^\circ}\right]$$

$$= \frac{4\left[\sin\left(60^\circ - 20^\circ\right)\right]}{\sin 40^\circ}$$

$= 4$

12. If $\sin\theta + \sin\phi = a$, $\cos\theta + \cos\phi = b$.

(i) Prove that $\tan\left(\frac{\theta+\phi}{2}\right) = \frac{a}{b}$ (ii) Prove that $\sin(\theta+\phi) = \frac{2ab}{a^2+b^2}$

(iii)Prove that $\cos(\theta+\phi) = \frac{b^2-a^2}{b^2+a^2}$ (iv) Prove that $\cos(\theta-\phi) = \frac{a^2+b^2-2}{2}$

Sol. (i) $\tan\left(\frac{\theta+\phi}{2}\right) = \frac{\sin\left(\frac{\theta+\phi}{2}\right)}{\cos\left(\frac{\theta+\phi}{2}\right)}$

$\sin\theta + \sin\phi = a$, $\cos\theta + \cos\phi = b$

$2\sin\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = a$ $a\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = b$

$$\frac{2\sin\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = a}{2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = b}$$

$$\tan\left(\frac{\theta+\phi}{2}\right) = \frac{a}{b}$$

(ii) $a^2 + b^2 = 2 + 2(\sin\theta\sin\phi + \cos\theta\cos\phi) \quad \Rightarrow 2[1+\cos(\theta-\phi)]$

$2ab = 2\sin(\theta+\phi) + 2\sin\theta\cos\theta + 2\sin\phi\cos\phi$

$= 2\sin(\theta+\phi) + \sin 2\theta + \sin 2\phi$

$= 2\sin(\theta+\phi) + 2\sin(\theta+\phi)\cos(\theta.\phi)$

$= 2\sin(\theta+\phi)\quad [1+\cos(\theta-\phi)]$

$$\frac{2ab}{a^2+b^2} = \frac{2\sin(\theta+\phi)[1+\cos(\theta-\phi)]}{2[1+\cos(\theta-\phi)]}$$

$= \sin(\theta+\phi)$

(iii) $b^2 - a^2 = \cos^2\theta - \sin^2\theta + \cos^2\phi - \sin^2\phi + 2(\cos\theta\cos\phi - \sin\theta\sin\phi)$

$= \cos 2\theta + \cos 2\phi + 2[\cos(\theta-\phi)]$

$= 2\cos(\theta+\phi)\cos(\theta-\phi) + 2\cos(\theta-\phi)$

$$\frac{b^2-a^2 = 2[\cos(\theta+\phi)[\cos(\theta-\phi)+1]}{b^2+a^2 = 2[1+\cos(\theta-\phi)]}$$

$$\frac{b^2-a^2}{b^2+a^2} = \cos(\theta+\phi)$$

(iv) $a^2 + b^2 - 2 = 2[1 + \cos(\theta - \phi)] - 2$

$$\frac{a^2 + b^2 - 2}{2} = \frac{2 + 2\cos(\theta - \phi) - 2}{x} = \cos(\theta - \phi)$$

13. Find value of

(i) $\sin 7.5^\circ$ (ii) $\cos 7.5^\circ$ (iii) $\tan 7.5^\circ$ (iv) $\cot 7.5^\circ$

Ans. (i) $\sin(7.5^\circ) = \frac{\sqrt{4-\sqrt{6}-\sqrt{2}}}{2\sqrt{2}}$ (ii) $\cos(7.5^\circ) = \frac{\sqrt{4+\sqrt{6}+\sqrt{2}}}{2\sqrt{2}}$

(iii) $\tan(7.5^\circ) = \frac{2\sqrt{2}-\sqrt{3}-1}{\sqrt{3}-1}$ (iv) $\cot(7.5^\circ) = \frac{2\sqrt{2}+\sqrt{3}+1}{\sqrt{3}-1}$

Sol. (i) $\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$ $\cos 15^\circ = \sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$

$$\sin 7.5 = \sqrt{\frac{1-\cos 15^\circ}{2}} \Rightarrow \sqrt{\frac{1-\frac{\sqrt{3}+1}{2\sqrt{2}}}{2}}$$

$$= \sqrt{\frac{2\sqrt{2}-\sqrt{3}-1}{4\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}} \Rightarrow \sqrt{\frac{4-\sqrt{6}-\sqrt{2}}{8}}$$

$$= \frac{\sqrt{4-\sqrt{6}-\sqrt{2}}}{2\sqrt{2}}$$

$$\text{(ii)} = \sqrt{\frac{1+\cos 15}{2}} \Rightarrow \sqrt{\frac{1+\frac{\sqrt{3}+1}{2\sqrt{2}}}{2}}$$

$$= \sqrt{\frac{4+\sqrt{6}+\sqrt{2}}{8}} \Rightarrow \frac{\sqrt{4+\sqrt{6}+\sqrt{2}}}{2\sqrt{2}}$$

$$\text{(iii)} \tan 7.5 = \frac{\sin 7.5}{\cos 7.5}$$

$$= \frac{\sqrt{4-\left(\sqrt{6}+\sqrt{2}\right)}}{\sqrt{4+\left(\sqrt{6}+\sqrt{2}\right)}} \times \sqrt{\frac{4-\left(\sqrt{6}+\sqrt{2}\right)}{4-\left(\sqrt{6}+\sqrt{2}\right)}}$$

$$= \frac{4-\sqrt{6}-\sqrt{2}}{\sqrt{16-8-4\sqrt{3}}} \Rightarrow \frac{4-\sqrt{6}-\sqrt{2}}{\sqrt{8-4\sqrt{3}}}$$

$$= \frac{4-\sqrt{6}-\sqrt{2}}{\sqrt{2}\sqrt{4-2\sqrt{3}}} \Rightarrow \frac{2\sqrt{2}-\sqrt{3}-1}{\sqrt{3}-1}$$

$$\text{(iv) } \cot 7.5 = \frac{\cos 7.5}{\sin 7.5} \Rightarrow \frac{\sqrt{4+\left(\sqrt{6}+\sqrt{2}\right)}}{\sqrt{4-\left(\sqrt{6}+\sqrt{2}\right)}}$$

$$= \frac{4+\sqrt{6}+\sqrt{2}}{\sqrt{16-8-2\sqrt{3}}} \Rightarrow \frac{2\sqrt{2}+\sqrt{3}+1}{\sqrt{3}-1}$$

14. Prove that $\tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A$.

Sol. $\tan 3A = \tan (2A + A)$

$$\tan 3A = \frac{\tan 2A + \tan A}{1 - \tan 2A \tan A}$$

$\tan 3A - \tan 3A \tan 2A \tan A = \tan 2A + \tan A$

$\tan 3A - \tan 2A - \tan A = \tan 3A \tan 2A \tan A$.

15. Prove that : $\tan 70^\circ = 2 \tan 50^\circ + \tan 20^\circ$ .

***Sol:*** $\tan 70^\circ = \tan (50^\circ + 20^\circ) = \dfrac{\tan 50 + \tan 20}{1 - \tan 50 \tan 20^\circ}$

$\Rightarrow \tan 70^\circ - \tan 70^\circ \tan 50^\circ + \tan 20^\circ = \tan 50^\circ + \tan 20^\circ$

$\Rightarrow \tan 70^\circ = \tan 70^\circ \tan 50^\circ \tan 20^\circ + \tan 50^\circ + \tan 20^\circ$

$= \tan 70^\circ \tan 50^\circ \cot 70^\circ + \tan 50^\circ + \tan 20^\circ$

$= 2 \tan 50^\circ + \tan 20^\circ$ .

16. If $\sin\theta + \cos\theta = \dfrac{1}{2}$, then $16\left(\sin\left(2\theta\right) + \cos\left(4\theta\right) + \sin\left(6\theta\right)\right)$ is equal to : **[JEE(Main)July2021]**

यदि $\sin\theta + \cos\theta = \dfrac{1}{2}$ है, तो $16\left(\sin\left(2\theta\right) + \cos\left(4\theta\right) + \sin\left(6\theta\right)\right)$ बराबर है : **[JEE(Main)July2021]**

**Ans.** – 23

**Sol.** $\sin\theta + \cos\theta = \dfrac{1}{2} \qquad \Rightarrow \sin 2\theta = \dfrac{-3}{4}$

$16 (\sin 2\theta + \cos 4\theta + \sin 6\theta)$

$= 16 (\sin 2\theta + (1 - 2 \sin^2 2\theta) + (3 \sin 2\theta - 4 \sin^3 2\theta)$

$= 16 (1 + 4 \sin 2\theta - 2 \sin^2 2\theta - 4 \sin^3 2\theta)$

$$16\left(1 + 4\left(\frac{-3}{4}\right) - 2\left(\frac{-3}{4}\right)^2 - 4\left(-\frac{3}{4}\right)^3\right)$$

$$= 16\left(1 - 3 - \frac{9}{8} + \frac{27}{16}\right)$$

$= -32 - 18 + 27$

$= -23$

## 3.3 3A FORMULAE

1. $\sin 3A = 3\sin A - 4\sin^3 A$
2. $\cos 3A = 4\cos^3 A - 3\cos A$
3. $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

## 3.4 6 IMPORTANT RESULTS

1. $\cot A - \tan A = 2\cot 2A$
2. $\cot A + \tan A = 2\operatorname{cosec} 2A$
3. $(1 + \tan A)(1 + \tan B) = 2$ if $A + B = 45^\circ$
4. $\sin\theta \sin(60^\circ - \theta)\sin(60^\circ + \theta) = \dfrac{1}{4}\sin 3\theta$
5. $\cos\theta . \cos(60^\circ - \theta)\cos(60^\circ + \theta) = \dfrac{1}{4}\cos 3\theta$
6. $\tan\theta \tan(60^\circ - \theta)\tan(60^\circ + \theta) = \tan 3\theta$

### Questions

Find value of

1. $\tan\dfrac{\pi}{24} + \tan\dfrac{5\pi}{24} + \tan\dfrac{7\pi}{24} + \tan\dfrac{11\pi}{24}$.

Ans. $4\sqrt{6}$

Sol. $\tan\dfrac{\pi}{24} + \tan\left(\dfrac{\pi}{2} - \dfrac{\pi}{24}\right) + \tan\dfrac{5\pi}{24} + \tan\left(\dfrac{\pi}{2} - \dfrac{5\pi}{24}\right)$

$\tan\dfrac{\pi}{24} + \cot\dfrac{\pi}{24} + \tan\dfrac{5\pi}{24} + \cot\dfrac{5\pi}{24}$

$2\operatorname{cosec}\dfrac{2\pi}{24} + 2\operatorname{cosec}\dfrac{2.5\pi}{24}$

$2\left(\operatorname{cosec}\dfrac{\pi}{12} + \operatorname{cosec}\dfrac{5\pi}{12}\right)$

$2(\operatorname{cosec} 15^\circ + \operatorname{cosec} 75^\circ)$

$2(\operatorname{cosec} 15^\circ + \sec 15^\circ)$

$2\left(\dfrac{1}{\sin 15^\circ} + \dfrac{1}{\cos 15^\circ}\right)$

$2\left(\dfrac{\cos 15^\circ + \sin 15^\circ}{\sin 15^\circ \cos 15^\circ}\right) \Rightarrow \dfrac{4\left(\dfrac{\sqrt{3}+1}{2\sqrt{2}} + \dfrac{\sqrt{3}-1}{2\sqrt{2}}\right)}{\sin 30^\circ}$

$$= \frac{4\left(\frac{\sqrt{3}}{\sqrt{2}}\right)}{\frac{1}{2}} \Rightarrow \frac{8\sqrt{3}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= 4\sqrt{6}$$

2. $\tan\frac{\pi}{24} - \tan\frac{5\pi}{24} + \tan\frac{7\pi}{24} - \tan\frac{11\pi}{24}$

Ans. $-4\sqrt{3}$

Sol. $\tan\frac{\pi}{24} - \cot\frac{\pi}{24} + \frac{\tan 7\pi}{24} - \cot\frac{7\pi}{24}$

$$= -2\cot\frac{\pi}{12} - 2\cot\frac{7\pi}{12}$$

$$= -2\left(\frac{1}{\tan 15^\circ} + \frac{1}{\tan 105^\circ}\right) \Rightarrow -2\left(\frac{1}{2-\sqrt{3}} - (2-\sqrt{3})\right)$$

$$= -2\left(\frac{1-(4+3-4\sqrt{3})}{2-\sqrt{3}}\right) \Rightarrow -2\left(\frac{1-7+4\sqrt{3}}{2-\sqrt{3}}\right)$$

$$= -2\left(\frac{-6+4\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}\right)$$

$$= -2(-12-6\sqrt{3}+8\sqrt{3}+12)$$

$$= -4\sqrt{3}$$

3. $\tan^2\frac{\pi}{24} + \tan^2\frac{5\pi}{24} + \tan^2\left(\frac{7\pi}{24}\right) + \tan^2\left(\frac{11\pi}{24}\right)$.

Ans. 60

Sol. $\tan^2\frac{\pi}{24} + \cot^2\frac{\pi}{24} + \tan^2\frac{5\pi}{24} + \cot^2\left(\frac{5\pi}{24}\right)$

$$\left(\tan\frac{\pi}{24} + \cot\frac{\pi}{24}\right)^2 - 2 + \left(\tan\frac{5\pi}{24} + \cot\frac{5\pi}{24}\right)^2 - 2$$

$$\left(2\operatorname{cosec}\frac{\pi}{12}\right)^2 + \left(2\operatorname{cosec}\frac{5\pi}{12}\right)^2 - 4$$

$$\left(\frac{2}{\sin 15^\circ}\right)^2 + \left(\frac{2}{\sin 75^\circ}\right)^2 - 4$$

$$\left(\frac{2}{\frac{\sqrt{3}-1}{2\sqrt{2}}}\right)^2+\left(\frac{2}{\frac{\sqrt{3}+1}{2\sqrt{2}}}\right)^2-4$$

$$\left(\frac{4\sqrt{2}}{\sqrt{3}-1}\right)^2+\left(\frac{4\sqrt{2}}{\sqrt{3}+1}\right)^2-4$$

$$32\left(\frac{1}{4-2\sqrt{3}}+\frac{1}{4+2\sqrt{3}}\right)-4$$

$$32\left(\frac{4+2\sqrt{3}+4-2\sqrt{3}}{16-12}\right)-4$$

$$\frac{32.8}{4}-4$$

$64 - 4 \Rightarrow 60$

4. $\prod_{r=1}^{8}\left(1+\tan\left(5r^{o}\right)\right)$

Ans. 16

Sol. $(1 + \tan5°).(1 + \tan10°)\,(1 + \tan15°)$.....$(1 + \tan40°)$
$(1 + \tan5°)\,(1 + \tan40°)\,(1 + \tan10°)\,(1 + \tan35°)\,(1 + \tan15°)$
$(1 + \tan30°)\,(1 + \tan20°)\,(1 + \tan25°)$
$2 \cdot 2 \cdot 2 \cdot 2 \Rightarrow 16$

5. $\sin 5^{o} \sin 55^{o} \sin 65^{o}$.

Ans. $\frac{\sqrt{3}-1}{8\sqrt{2}}$

Sol. $\sin5\,(\sin60 - 5)\,(\sin\,(\sin\,(60 + 5))$

$$\frac{1}{4}\sin 3(5) \Rightarrow \frac{1}{4}\sin 15^{\circ} \Rightarrow \frac{1}{4}\frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$=\frac{\sqrt{3}-1}{8\sqrt{2}}$$

6. $\cos 12^{o} \cos 24^{o}, \cos 48^{o} \cos 84^{o}$

Ans. $\frac{1}{16}$

Sol. $$\frac{1}{2\sin 12^{\circ}} 2\sin 12^{\circ} \cos 12^{\circ} \cos 24^{\circ} \cos 48^{\circ} \cos 84^{\circ}$$

$$\frac{1}{2^2 \sin 12^\circ} 2\sin 24^\circ \cos 24^\circ \cos 48^\circ \cos 84^\circ$$

$$\frac{1}{2^3 \sin 12^\circ} 2\sin 48^\circ \cos 48^\circ \cos 84^\circ$$

$$\frac{1}{2^3 \text{Sin}12^\circ} \sin 96^\circ \cos 84^\circ \qquad \Rightarrow \frac{\sin(90+6)}{2^3 \text{Sin}12} \cos(90-6)$$

$$= \frac{2\cos 6 \sin 6}{2^4 \sin 12} \Rightarrow \frac{\sin 12}{16 \sin 12} \Rightarrow \frac{1}{16}$$

7. $f(x) = \tan^2 x + \cot^2 (x)$ find range of $f(x)$ if $x \in \left(0, \frac{\pi}{2}\right)$.

Ans. $[2, \infty)$

Sol. $= (\tan x - \cot x)^2 + 2$

$(\tan x - \cot x)^2 \geq 0$

$(\tan x - \cot x)^2 + 2 \geq 2$

$[2, \infty)$

**When to apply $\theta, 60 - \theta, 60 + \theta$, formulae**

Following points must be observed

(i) sum of two angles is $60^o$ (ii) sum of two angles is $120^o$ (iii) difference of two angles is $60^o$

## 3.5 VALUE AT 18 AND 36 DEGREES

$\theta = 18^o \Rightarrow 5\theta = 90^o$

$3\theta = \frac{\pi}{2} - 2\theta$

$\sin 3\theta = \sin\left(\frac{\pi}{2} - 2\theta\right) \Rightarrow \sin 3\theta = \cos 2\theta$

$3\sin\theta - 4\sin^3\theta = 1 - 2\sin^2\theta$

$4\sin^3\theta - 2\sin^2\theta - 3\sin\theta + 1 = 0$

$(\sin\theta - 1)(4\sin^2\theta + 2\sin\theta - 1) = 0$

$\sin\theta = 1$ (Rejected) or

$4\sin^2\theta + 2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{-1+\sqrt{5}}{4}$ or $\sin\theta = \frac{-1-\sqrt{5}}{4}$ (Rejected)

$\sin\frac{\pi}{10}$ or $\sin 18^\circ = \frac{\sqrt{5}-1}{4} = \cos 72^\circ$

$\cos 36^\circ$ or $\cos\frac{\pi}{5} = \frac{\sqrt{5}+1}{4} = \sin 54^\circ$

## 4.1 COMPONENDO AND DIVIDENDO

If $\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a+b}{a-b}=\frac{c+d}{c-d}$

Proof: – $\frac{a}{b}=\frac{c}{d}$

$\frac{a}{b}+1=\frac{c}{d}+1$ $\qquad$ $\frac{a}{b}-1=\frac{c}{d}-1$

$\frac{a+b}{b}=\frac{c+d}{a}$ .....(1) $\qquad$ $\frac{a-b}{b}=\frac{c-d}{a}$ .....(2)

$(1) \div (2)$

$\frac{a+b}{a-b}=\frac{c+d}{c-d}$

Q.1 If $\sin 2A = \lambda \sin 2B$, then prove that $\frac{\tan(A+B)}{\tan(A-B)}=\frac{\lambda+1}{\lambda-1}$

***Sol*** Given $\sin 2A = \lambda \sin 2B$

$\Rightarrow \quad \frac{\sin 2A}{\sin 2B}=\frac{\lambda}{1}$

Applying componendo & dividendo,

$\frac{\sin 2A+\sin 2B}{\sin 2B-\sin 2A}=\frac{\lambda+1}{1-\lambda}$

$\Rightarrow \quad \frac{2\sin\left(\frac{2A+2B}{2}\right)\cos\left(\frac{2A-2B}{2}\right)}{2\cos\left(\frac{2B+2A}{2}\right)\sin\left(\frac{2B-2A}{2}\right)}=\frac{\lambda+1}{1-\lambda}$

$\Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\sin\{-(A-B)\}}=\frac{\lambda+1}{1-\lambda}$ $\qquad \Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\times -\sin(A-B)}=\frac{\lambda+1}{-(\lambda-1)}$

$\Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\sin(A-B)}=\frac{\lambda+1}{\lambda-1}$ $\qquad \Rightarrow \quad \tan(A+B)\cot(A-B)=\frac{\lambda+1}{\lambda-1}$

$\Rightarrow \quad \frac{\tan(A+B)}{\tan(A-B)}=\frac{\lambda+1}{\lambda-1}$

**When to apply componendo & dividendo**

When ratio in sin/cos is given and ratio in tan/cot is asked and vice versa.

## 4.2 GENERAL FORMULA FOR TAN

$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$

$$\tan(A+B+C)=\frac{\tan A+\tan B+\tan C-\tan A\tan B\tan C}{1-\tan A\tan B-\tan B\tan C-\tan C\tan A}.$$

$$\tan(\theta_1+\theta_2+\theta_3+.....+\theta_n)=\frac{S_1-S_3+S_5-.....}{1-S_2+S_4-.....}$$

where $S_i$ denotes sum of product of tangent of angles taken i at a time

## 4.3 CONTINUED PRODUCT

$$\cos\theta\cdot\cos 2\theta\cdot\cos 2^2\theta\cdot\cos 2^3\theta.....\cos 2^{n-1}\theta=\frac{\sin 2^n\theta}{2^n\sin\theta}$$

**Questions**

Find value of

1. $\cos 36^\circ.\cos 72^\circ.$

Ans. $\frac{1}{4}$

Sol. $\frac{2\sin 36^\circ\cos 36^\circ\cos 72^\circ}{2\sin 36^\circ}$

$\frac{2\sin 72^\circ\cos 72^\circ}{4\sin 36^\circ}=\frac{\sin 144^\circ}{4\sin 36^\circ}$ $\Rightarrow\frac{\sin(180^\circ-36^\circ)}{4\sin 36^\circ}$

$=\frac{\sin 36^\circ}{4\sin 36^\circ}=\frac{1}{4}$

2. $\cos x\cdot\cos 2x.\cos 4x.....\cos 64x.$

Ans. $\frac{\sin(128x)}{128\sin x}$

Sol. $\cos x\cdot\cos 2x\cos 2^2x.....\cos 2^6x=\frac{\sin 2^7x}{2^7\sin x}$

$=\frac{\sin 128x}{128\sin x}$

3. $\cos x\cos\frac{x}{2}\cos\frac{x}{4}\cos\frac{x}{8}...\cos\frac{x}{32}$

Ans. $\frac{\sin 2x}{2^6\sin\left(\frac{x}{32}\right)}$

Sol. $\cos\frac{x}{2^5}\cos\frac{x}{2^4}.....\cos x$

$\frac{\sin 2^6\left(\frac{x}{2^5}\right)}{2^6\sin\left(\frac{x}{2^5}\right)}\Rightarrow\frac{\sin 2x}{2^6\sin\frac{x}{32}}$

4. $\cos 20^\circ \cos 40^\circ \cos 80^\circ$.

Ans. $\frac{1}{8}$

Sol. $\frac{\sin 2^3(20)}{2^3 \sin 20^\circ} \Rightarrow \frac{\sin 160^\circ}{8 \sin 20^\circ}$

$= \frac{1}{8}$

5. $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$

Ans. $\frac{1}{8}$

Sol. $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\left(\pi - \frac{4\pi}{7}\right)$

$-\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}$

$-\frac{-\sin 2^3\left(\frac{\pi}{7}\right)}{2^3 \sin\frac{\pi}{7}} \Rightarrow \frac{\sin\frac{8\pi}{7}}{8\sin\frac{\pi}{7}} \Rightarrow \frac{-\sin\left(\pi + \frac{\pi}{7}\right)}{8\sin\frac{\pi}{7}}$

$= \frac{\sin\frac{\pi}{7}}{8\sin\frac{\pi}{7}} = \frac{1}{8}$

6. $\cos 12^\circ \cos 24^\circ \cos 48^\circ . \cos 84^\circ$

Ans. $\frac{1}{16}$

Sol. $\frac{\sin 2^3(12)}{2^3 \sin 12}\cos 84 \Rightarrow \frac{\sin 96 \cos 84}{8 \sin 12}$

$\frac{2\cos 6 \sin 6}{16 \sin 12} \Rightarrow \frac{1}{16}$

7. $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}$.

Ans. $\frac{1}{8}$

Sol. $\sin\left(\frac{\pi}{2} - \frac{6\pi}{14}\right)\sin\left(\frac{\pi}{2} - \frac{4\pi}{14}\right)\sin\left(\frac{\pi}{2} - \frac{2\pi}{14}\right)$

$\cos\frac{6\pi}{14}\cos\frac{4\pi}{14}\cos\frac{2\pi}{14}$

$\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$

8. $\prod_{r=1}^{7} \sin\left((2r-1)\frac{\pi}{14}\right)$

Ans. $\frac{1}{64}$

Sol. $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\cdots\sin\frac{13\pi}{14}$ $\qquad \frac{\pi}{2}+x=\frac{9\pi}{14} \qquad x=\frac{9\pi}{14}-\frac{\pi}{2}$

$\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}\sin\frac{\pi}{2}\sin\left(\frac{\pi}{2}+\frac{2\pi}{14}\right)\sin\left(\frac{\pi}{2}+\frac{4\pi}{14}\right)$

$\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}\sin\frac{\pi}{2}\sin\left(\frac{\pi}{2}+\frac{2\pi}{14}\right)\sin\left(\frac{\pi}{2}+\frac{4\pi}{14}\right)\sin\left(\frac{\pi}{2}+\frac{6\pi}{14}\right)$

$\underbrace{\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}}\underbrace{\cos\frac{\pi}{7}\cos\frac{8\pi}{7}\cos\frac{3\pi}{7}}$

$\frac{1}{8}\times\frac{1}{8}=\frac{1}{64}$

9. $\prod_{r=1}^{4} \sin\left((2r-1)\frac{\pi}{8}\right)$

Ans. $\frac{1}{8}$

Sol. $\sin\frac{\pi}{8}\sin\frac{3\pi}{8}\sin\frac{5\pi}{8}\sin\frac{7\pi}{8}$ $\qquad \frac{\pi}{2}-\frac{\pi}{8}$

$\sin\frac{\pi}{8}\sin\frac{3\pi}{8}\sin\left(\pi-\frac{3\pi}{8}\right)\sin\left(\pi-\frac{\pi}{8}\right)$

$\sin\frac{\pi}{8}\sin\frac{3\pi}{8}\sin\frac{3\pi}{8}\sin\frac{\pi}{8}$

$\sin^2\left(\frac{\pi}{2}-\frac{3\pi}{8}\right)\sin^2\frac{3\pi}{8}$

$\cos^2\frac{3\pi}{8}\sin^2\frac{3\pi}{8}$

$\frac{4}{4}\sin^2\frac{3\pi}{8}\cos^2\frac{3\pi}{8}$

$\frac{1}{4}\left(2\sin\frac{3\pi}{8}\cos\frac{3\pi}{8}\right)^2 \Rightarrow \frac{1}{4}\left(\sin\frac{3\pi}{4}\right)^2$

$=\frac{1}{4}\left(\frac{1}{\sqrt{2}}\right)^2 \Rightarrow \frac{1}{4}\times\frac{1}{2}\Rightarrow\frac{1}{8}$

## 5. TRIGONOMETRIC IDENTITIES IN A TRIANGLE

If A, B, C denote the angle of a triangle ABC, then the relation $A + B + C = \pi$ enables us to establish many important identities involving trigonometric ratios of these angles.

(I) If $A + B + C = \pi$, then $A + B = \pi - C$, $B + C = \pi - A$ and $C + A = \pi - B$

$$\sin(A + B) = \sin(\pi - C) = \sin C$$
$$\cos(A + B) = \cos(\pi - C) = -\cos C$$
$$\tan(A + B) = \tan(\pi - C) = -\tan C$$

(II) If $A + B + C = \pi$, then $\frac{A+B}{2} = \frac{\pi}{2} - \frac{C}{2}$ and $\frac{B+C}{2} = \frac{\pi}{2} - \frac{A}{2}$ and $\frac{C+A}{2} = \frac{\pi}{2} - \frac{B}{2}$

$$\sin\left(\frac{A+B}{2}\right) = \sin\left(\frac{\pi}{2} - \frac{C}{2}\right) = \cos\left(\frac{C}{2}\right); \cos\left(\frac{A+B}{2}\right) = \cos\left(\frac{\pi}{2} - \frac{C}{2}\right) = \sin\left(\frac{C}{2}\right); \tan\left(\frac{A+B}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right) = \cot\left(\frac{C}{2}\right)$$

**Questions**

Prove that

1. $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$.

Sol $\sin 2A + \sin 2B + \sin 2C$

$= 2\sin(A + B).\cos(A - B) + \sin 2C$

$= 2 \sin(\pi - C).\cos(A - B) + \sin 2C$

$= 2\sin C \cos(A - B) + 2 \sin C \cos C$

$= 2 \sin C [\cos(A - B) + \cos C]$

$= 2 \sin C [\cos(A - B) - \cos(A + B)]$

$= 2 \sin C [2\sin A \sin B]$

$= 4 \sin A \sin B \sin C$

2. $\sin 2A + \sin 2B - \sin 2C = 4 \cos A \cos B \sin C$.

Sol. $= 2\sin A \cos A + 2\cos(B + C) \sin(B - C)$

$= 2\sin A \cos A - 2\cos A \sin(B - C)$

$= 2\cos A [\sin A - \sin(B - C)]$

$= 2\cos A[\sin(B + C) - \sin(B - C)]$

$= 2\cos A[2\cos B \sin C]$

$= 4\cos A \cos B \sin C$.

3. $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$.

Sol. $\cos 2A + \cos 2B + \cos 2C = 2 \cos(A + B) \cos(A - B) + \cos 2C$

$= 2 \cos(\pi - C) \cos(A - B) + \cos 2C \quad \{\because A + B + C = \pi\}$

$= -2 \cos C \cos(A - B) + 2 \cos^2 C - 1$

$= -1 + 2 \cos C (\cos C - \cos(A - B))$

$= -1 + 2 \cos C (-\cos(A + B) - \cos(A - B))$

$= -1 - 4\cos A\cos B\cos C$

4. $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}.$

Sol. $\sin A + \sin B + \sin C = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$= 2\cos\frac{C}{2}\cos\left(\frac{A-B}{2}\right) + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$= 2\cos\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) + \cos\left(\frac{A+B}{2}\right)\right]$

$= 2\cos\frac{C}{2}\left[2\cos\frac{A}{2}\cos\frac{B}{2}\right]$

$= 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

5. $\sin^2 A + \sin^2 B - \sin^2 C = 2\sin A\sin B\cos C.$

Sol. $= \sin^2 A + \sin(B+C).\sin(B-C)$

$= \sin^2 A + \sin A.\sin(B-C)$

$= \sin A[\sin A + \sin(B-C)]$

$= \sin A[\sin(B+C) + \sin(B-C)]$

$= \sin A[2\sin B\cos C]$

$= 2\sin A\sin B\cos C.$

6. $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2\cos A\cos B\cos C.$

Sol $S = \sin^2 A + \sin^2 B + \sin^2 C$

so that $2S = 2\sin^2 A + 1 - \cos 2B + 1 - \cos 2C$

$= 2\sin^2 A + 2 - 2\cos(B+C)\cos(B-C)$

$= 2 - 2\cos^2 A + 2 - 2\cos(B+C)\cos(B-C)$

$\therefore$ $S = 2 + \cos A[\cos(B-C) + \cos(B+C)]$

since $\cos A = -\cos(B+C)$

$\therefore$ $S = 2 + 2\cos A\cos B\cos C$

7. $\tan A + \tan B + \tan C = \tan A\tan B\tan C.$

Sol. $\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - \tan A\tan B - \tan B\tan C - \tan C\tan A}$

$\tan\pi = \frac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - (\tan A\tan B + \tan B\tan C + \tan C\tan A)}$

$O = \frac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - (\tan A\tan B + \tan B\tan C + \tan C\tan A)}$

$O = \tan A + \tan B + \tan C - \tan A\tan B\tan C$

$\tan A \tan B \tan C = \tan A + \tan B + \tan C$

8. $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

Sol. $\tan A \tan B \tan C = \tan A + \tan B + \tan C$

$$\frac{1}{\cot A \cot B \cot C} = \frac{1}{\cot A} + \frac{1}{\cot B} + \frac{1}{\cot C}$$

$$\frac{1}{\cot A \cot B \cot C} = \frac{\cot B \cot C + \cot A \cot C + \cot A \cot B}{\cot A \cot B \cot C}$$

$$\frac{\cot A \cot B \cot C}{\cot A \cot B \cot C} = \cot A \cot B + \cot B \cot C + \cot C \cot A$$

$1 = \cot A \cot B + \cot B \cot C + \cot C \cot A$

9. $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$

Sol. $$\tan\left(\frac{A}{2} + \frac{B}{2} + \frac{C}{2}\right) = \frac{\tan\left(\frac{A}{2}\right) + \tan\frac{B}{2} + \tan\frac{C}{2} - \tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2}}{1 - \left(\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2}\right)}$$

$$\frac{1}{0} = \frac{\tan\left(\frac{A}{2}\right) + \tan\frac{B}{2} + \tan\frac{C}{2} - \tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2}}{1 - \left(\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2}\right)}$$

$$1 - \left(\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2}\right) = 0$$

$$1 = \tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2}$$

10. $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$.

Note

If $\sin^2 A + \sin^2 B + \sin^2 C = 2$ or $\cos^2 A + \cos^2 B + \cos^2 C = 1$ then triangle is right angled.

Sol. $$\cot\left(\frac{A}{2} + \frac{B}{2}\right) = \cot\left(\frac{\pi}{2} - \frac{C}{2}\right) \Rightarrow \tan\frac{C}{2}$$

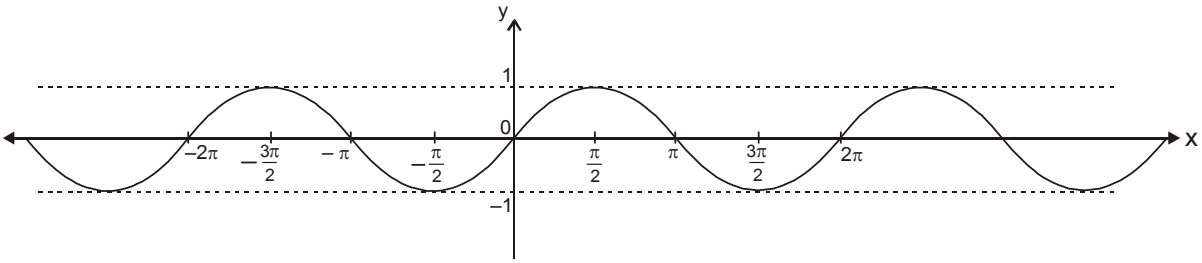
$$\frac{\cot\frac{A}{2}\cot\frac{B}{2} - 1}{\cot\frac{A}{2} + \cot\frac{B}{2}} = \frac{1}{\cot\frac{C}{2}}$$

$$\cot\frac{C}{2}\cot\frac{A}{2}\cot\frac{B}{2} - \cot\frac{C}{2} = \cot\frac{A}{2} + \cot\frac{B}{2}$$

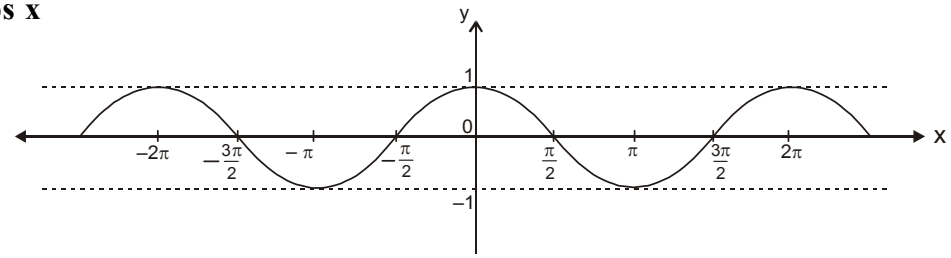
$$\cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}=\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}$$

## 6. GRAPHS OF TRIGONOMETRIC FUNCTION

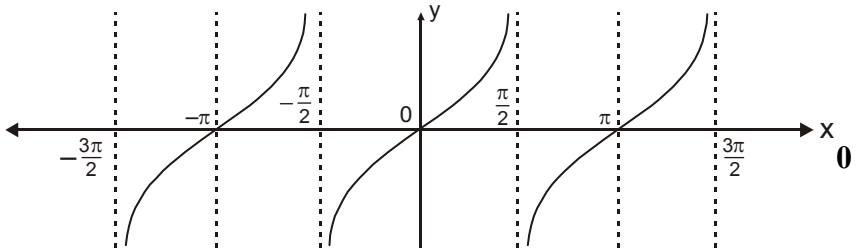
**(i) y = sin x**



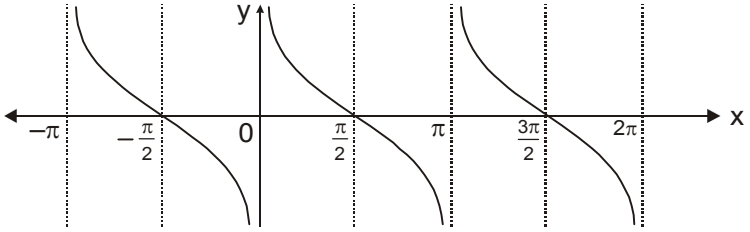

**(ii) y = cos x**



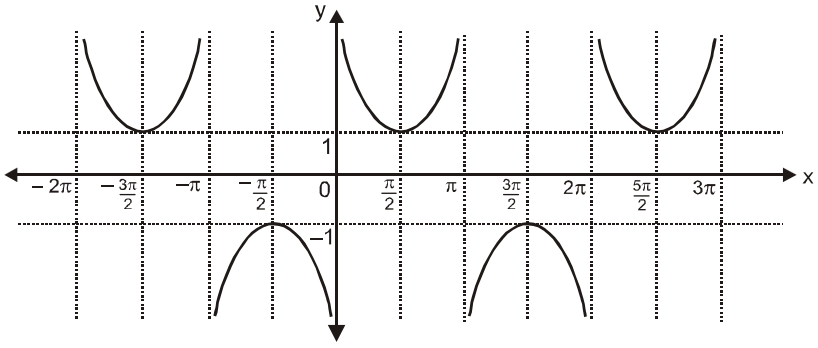

**(iii) y = tan x**



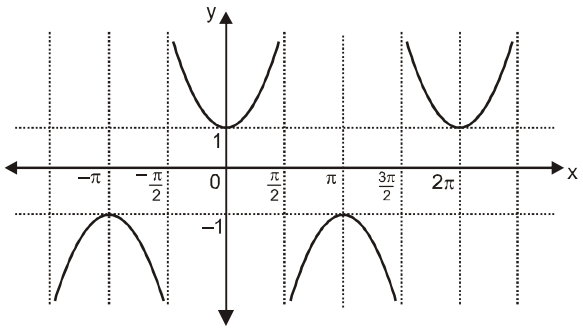

**(iv) y = cot x**




**(v) y = cosec x**



**(vi) y = sec x**




## 7.1 RANGE OF TRIGONOMETRIC FUNCTIONS TYPE - 1

$\left.\begin{matrix}\sin x\\ \cos x\end{matrix}\right]\in[-1,1]$ $\qquad$ $\left.\begin{matrix}\sin^2 x\\ \cos^2 x\end{matrix}\right]\in[0,1]$

$\left.\begin{matrix}\tan x\\ \cot x\end{matrix}\right]\in(-\infty,\infty)$ $\qquad$ $\left.\begin{matrix}\tan^2 x\\ \cot^2 x\end{matrix}\right]\in[0,\infty)$

$\left.\begin{matrix}\sec x\\ \operatorname{cosec} x\end{matrix}\right]\in(-\infty,-1]\cup[1,\infty)$ $\qquad$ $\left.\begin{matrix}\sec^2 x\\ \operatorname{cosec}^2 x\end{matrix}\right]\in[1,\infty)$

**Questions**

Find range of y

1. $y = 2\sin x - 3$

Ans. $[-5, -1]$

Sol. $-1 \le \sin x \le 1$

$-2 \le 2\sin x \le 2$

$-2 - 3 \le 2\sin x - 3 \le 2 - 3$

$-5 \le 2\sin x - 3 \le -1$

$y \in [-5, -1]$

2. $y = 3 - 4\cos x$

Ans. $[-1, 7]$

Sol. $-1 \le \cos x \le 1$

$-4 \le 4\cos x \le 4$

$4 \ge -4\cos x \ge -4$

$7 \ge 3 - 4\cos x \ge -1$

$y \in [-1, 7]$

3. $y = 3\tan^2 x - 5$

Ans. $[-5, \infty)$

Sol. $0 \le \tan^2 x < \infty$

$0 \le 3\tan^2 x < \infty$

$-5 \le 3\tan^2 x - 5 < \infty$

$y \in [-5, \infty]$

4. $y = 2 - 3\sec^2 x$

Ans. $(-\infty, -1]$

Sol. $1 \le \sec^2 x < \infty$

$3 \le 3\sec^2 x < \infty$

$-3 \ge -3\sec^2 x > -\infty$

$-1 \ge 2 - 3\sec^2 x > -\infty$

$y \in [-\infty, -1]$

## 7.2 RANGE OF TRIGONOMETRIC FUNCTIONS TYPE - 2

quadratic in sin or cos

Find range of y

1. $y = \sin^2 x - 4\sin x + 3$

Ans. $[0, 8]$

Sol. $y = \sin^2 x - 4\sin x + 4 - 1$

$y = (\sin x - 2)^2 - 1$

$-1 \le \sin x \le 1$

$-3 \le \sin x - 2 \le -1$

$9 \ge (\sin x - 2)^2 \ge 1$

$8 \ge (\sin x - 2)^2 - 1 \ge 0$

$y \in [0, 8]$

2. $y = \cos^2 x - 4\sin x + 3$

Ans. $[-1, 7]$

Sol. $y = 1 - \sin^2 x - 4\sin x + 3$

$y = -\sin^2 x - 4\sin x + 4$

$y = -(\sin x + 2)^2 + 8$

$-1 \le \sin x \le 1$

$1 \le \sin x + 2 \le 3$

$1 \le (\sin x + 2)^2 \le 9$

$-1 \ge -(\sin x + 2)^2 \ge -9$

$-1 + 8 \ge -(\sin x + 2)^2 + 8 \ge -1$

$7 \ge -(\sin x + 2)^2 + 7 \ge -1$

$y \in [-1, 7]$

3. $y = \sin^2 x - 6 \cos x + 4$

Ans. $[-2, 10]$

Sol. $y = 1 - \cos^2 x - 6\cos x + 4$

$y = -\cos^2 x - 6\cos x + 5 + 9 - 9$

$y = -(\cos x + 3)^2 + 14$

$-1 \le \cos x + 3 \le 4$

$2 \le \cos x + 3 \le 4$

$4 \le (\cos x + 3)^2 \le 16$

$-4 \ge -(\cos x + 3)^2 \ge -16$

$-4 + 14 \ge -(\cos x + 3)^2 + 14 \ge -16 + 14$

$10 \ge -(\cos x + 3)^2 + 14 \ge -2$

$y \in [-2, 10]$

4. $y = \cos 2x + 4 \sin x + 2$

Ans. $[-3, 5]$

Sol.

5. $y = \cos 2x + 8 \cos x + 1$

Ans. $[-6, 10]$

Sol. $y = 2\cos^2 x - 1 + 8\cos x + 1$

$y = 2\cos^2 x + 8\cos x$

$= 2(\cos^2 x + 4\cos x + 4) - 8$

$= 2(\cos x + 2)^2 - 8$

$[-1, 1]$

$y_{min} = 2(-1 + 2)^2 - 8 = -6$

$y_{max} = 2(1 + 2)^2 - 8 = 10$

$y \in [-6, 10]$

6. $y=\dfrac{2\sin x-1}{\sin x-3}$ **[O]**

Ans. $\left[-\dfrac{1}{2},\dfrac{3}{4}\right]$

Sol. $y_{max}=\dfrac{2(-1)-1}{-1-3}\Rightarrow\dfrac{-3}{-4}\Rightarrow\dfrac{3}{4}$

$y_{min}=\dfrac{2(1)-1}{1-3}\Rightarrow\dfrac{1}{-2}\Rightarrow-\dfrac{1}{2}$

$y\in\left[-\dfrac{1}{2},\dfrac{3}{4}\right]$

7. $y=\dfrac{(2\sin x-1)^2+3}{4-3(2\sin x-1)^2}$ **[O]**

Ans. $\left(-\infty,\dfrac{-12}{23}\right]\cup\left[\dfrac{3}{4},\infty\right)$

Sol. Let $(2\sin x-1)^2=t \quad t\in[0,9]$

$y=\dfrac{t+3}{4-3t}$

$4y-3yt-t-3=0$

$-t(3y+1)=-(4y-3)$

$t=\dfrac{4y-3}{3y+1}$

$0\le\dfrac{4y-3}{3y+1}\le 9$

$0\le\dfrac{4y-3}{3y+1}$ $\qquad$ $\dfrac{4y-3}{3y+1}\le 9$

$0\le\dfrac{4y-3}{3y+1}$ $\qquad$ $\dfrac{4y-3-9(3y+1)}{3y+1}\le 0$

$\dfrac{4y-3-27y-9}{3y+1}\le 0$

$+\quad -\quad +$ $\qquad$ $\dfrac{-23y-12}{3y+1}\le 0$

$-1/3 \qquad 3/4$

$\left(-\infty,-\dfrac{1}{3}\right)\cup\left[\dfrac{3}{4},\infty\right)$ $\qquad$ $\dfrac{23y+12}{3y+1}\ge 0$

$+\quad -\quad +$

$-12/23 \qquad -1/3$

$$\left(-\infty,-\frac{12}{23}\right]\cup\left(-\frac{1}{3},\infty\right)$$

$$\left(-\infty,-\frac{12}{23}\right]\cup\left[\frac{3}{4},\infty\right)$$

## 7.3 RANGE OF TRIGONOMETRIC FUNCTIONS TYPE - 3

$$E = a\sin\theta + b\cos\theta$$

$$\Rightarrow \quad E = \sqrt{a^2+b^2}\left\{\frac{a}{\sqrt{a^2+b^2}}\sin\theta+\frac{b}{\sqrt{a^2+b^2}}\cos\theta\right\}$$

Let $\frac{b}{\sqrt{a^2+b^2}} = \sin\alpha$ & $\frac{a}{\sqrt{a^2+b^2}} = \cos\alpha$

$$\Rightarrow \quad E = \sqrt{a^2+b^2}\sin(\theta+\alpha), \text{ where } \tan\alpha = \frac{b}{a}$$

Hence for any real value of $\theta$,

$$-\sqrt{a^2+b^2} \le E \le \sqrt{a^2+b^2}$$

Find range of y

1. $y = \sin x + \cos x + 2$

Ans. $\left[2-\sqrt{2},2+\sqrt{2}\right]$

Sol. $-\sqrt{2} \le \sin x + \cos x \le \sqrt{2}$

$2-\sqrt{2} \le \sin x + \cos x + 2 \le 2+\sqrt{2}$

2. $y = \sin\theta + \sin\phi$.

Ans. $[-2, 2]$

Sol. $-1 \le \sin\theta \le 1 \quad -1 \le \sin\phi \le 1$

$y_{min} = -1 - 1 = -2$

$y_{max} = 1 + 1 = 2$

$[-2, 2]$

3. $y = 2\sin 2x - \cos 2x$.

Ans. $\left[-\sqrt{5},\sqrt{5}\right]$

Sol. $\left[-\sqrt{4+1},\sqrt{4+1}\right] \Rightarrow \left[-\sqrt{5},\sqrt{5}\right]$

4. $y = a\cos^2 x + b\sin^2 x + c\sin x\cos x + d$.

Ans. $= a\left(\frac{1+\cos 2x}{2}\right)+b\left(\frac{1-\cos 2x}{2}\right)+c\left(\frac{\sin 2x}{2}\right)+d$

$$= \frac{a}{2} + \frac{a\cos 2x}{2} + \frac{b}{2} - b\frac{\cos 2x}{2} + \frac{c}{2}\sin 2x + d$$

$$= \left(\frac{a}{2} - \frac{b}{2}\right)\cos 2x + \frac{c}{2}\sin 2x + \frac{a+b}{2} + d$$

$$y_{max} = \frac{a+b}{2} + d + \sqrt{\left(\frac{a-b}{2}\right)^2 + \frac{c^2}{4}}$$

$$y_{min} = \frac{a+b}{2} + d - \sqrt{\left(\frac{a-b}{2}\right)^2 + \frac{c^2}{4}}.$$

5. $y = \sqrt{3}\sin x + \cos x.$

Ans. $[-2, 2]$

Sol. $[-2, 2]$

6. Prove that : $-4 \le 5\cos\theta + 3\cos\left(\theta + \frac{\pi}{3}\right) + 3 \le 10$ , for all values of $\theta$. **[O]**

Sol. $5\cos\theta + 3\cos\left(\theta + \frac{\pi}{3}\right) + 3$

$$= 5\cos\theta + 3\cos\theta\cos\frac{\pi}{3} - 3\sin\theta\sin\frac{\pi}{3} + 3$$

$$= 5\cos\theta + \frac{3}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta + 3$$

$$= \frac{13\cos\theta}{2} - \frac{3\sqrt{3}}{2}\sin\theta + 3$$

$$y_{mini} = 3 - \sqrt{\frac{169}{4} + \frac{27}{4}}$$

$$= 3 - \sqrt{\frac{196}{4}} \Rightarrow 3 - \frac{14}{2} \Rightarrow -4$$

$y_{max} = 3 + 7 \Rightarrow 10$

7. Find the maximum value of $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2\cos\left(\frac{\pi}{4} - \theta\right)$ - **[O]**

**Ans 4**

**Sol** We have $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2\cos\left(\frac{\pi}{4} - \theta\right)$

$$= 1 + \frac{1}{\sqrt{2}}(\cos\theta + \sin\theta) + \sqrt{2}\ (\cos\theta + \sin\theta) = 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)(\cos\theta + \sin\theta)$$

$$= 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) \cdot \sqrt{2}\cos\left(\theta - \frac{\pi}{4}\right)$$

$$\therefore \text{maximum value} = 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) \cdot \sqrt{2} = 4$$

## 7.4 RANGE OF TRIGONOMETRIC FUNCTIONS TYPE - 4

1. $y = a^2 \tan^2 x + b^2 \cot^2 x \qquad a, b > 0$

$y = (a \tan x - b \cot x)^2 + 2ab$

check whether $a \tan x - b \cot x = 0$

$\tan^2 x = b/a$

$y|_{min} = 2ab$

$y \in [2ab, \infty)$

Q. $y = 9\tan^2 x + 16\cot^2 x$

Ans. $[24, \infty)$

Sol. $y = (3\tan x - 4\cot x)^2 - 2(3)(4)$

$y = (3\tan x - 4\cot x)^2 - 24$

$3\tan x = 4\cot x$

$\tan^2 x = \frac{4}{3}$

$\tan x = \frac{2}{\sqrt{3}}$

$y_{min} = 24$

$y \in [24, \infty)$

2. $y = a^2 \sec^2 x + b^2 \operatorname{cosec}^2 x$

Sol. $= a^2(1 + \tan^2 x) + b^2(1 + \cot^2 x)$

$= (a^2 + b^2) + (a^2 \tan^2 x + b^2 \cot^2 x)$

$= (a^2 + b^2) + [2ab, \infty)$

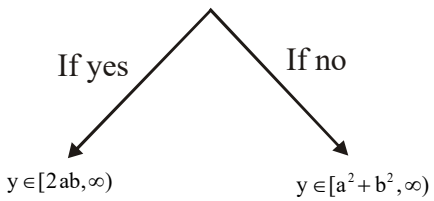
$y \in [(a+b)^2, \infty)$

3. $y = a^2 \sin^2 x + b^2 \operatorname{cosec}^2 x$

$= (a \sin x - b \operatorname{cosec} x)^2 + 2ab$

check whether $a \sin x = b \operatorname{cosec} x$

$\sin^2 x = b/a$




Ex:. $y = 25\sin^2 x + 9\operatorname{cosec}^2 x$

$25\sin^2 x = 9\operatorname{cosec}^2 x$

$\sin^4 x = \frac{9}{25}$

$y \in [30, \infty)$

Ex. $y = 9\sin^2 x + 25\operatorname{cosec}^2 x$

$9\sin^2 x = 25\operatorname{cosec}^2 x$

$\sin^4 x = \frac{25}{9}$

$y = [9 + 25, \infty)$

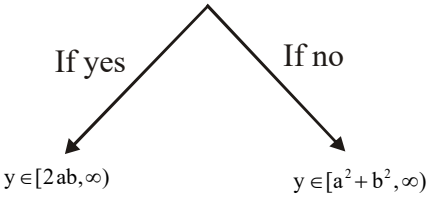
$$\Rightarrow y = 9\sin^2 x + 25\operatorname{cosec}^2 x$$

$$= 9\left(\underbrace{\sin^2 x + \frac{1}{\sin^2 x}}_{\min=2}\right) + 16\underbrace{\operatorname{cosec}^2 x}_{\min=1}$$

4. $y = a^2\cos^2 x + b^2\sec^2 x$

check whether $a\cos x = b\sec x$

$$\cos^2 x = b/a$$




## 8.1 SUMMATION OF SERIES TYPE - 1

(i) $\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin\{\alpha + (n-1)\beta\} = \dfrac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\sin\left(\alpha + \dfrac{n-1}{2}\beta\right)$

Sol. $S = \sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin\{\alpha + (n-1)\beta\}$

Multiply by $2\sin\dfrac{\beta}{2}$

$$2\sin\frac{\beta}{2}S = 2\sin\frac{\beta}{2}\sin\alpha + 2\sin\frac{\beta}{2}\sin(\alpha+\beta) + 2\sin\frac{\beta}{2}\sin(\alpha+2\beta)\ldots\ldots\ldots 2\sin\frac{\beta}{2}\sin\{\alpha+(n-1)\beta\}$$

$$2\sin\frac{\beta}{2}S = \cos\left(\alpha-\frac{\beta}{2}\right) - \cos\left(\alpha+\frac{\beta}{2}\right) + \cos\left(\alpha+\frac{\beta}{2}\right) - \cos\left(\alpha+\frac{\beta}{2}\right) +$$

$$\cos\left(\alpha+\frac{3\beta}{2}\right) - \cos\left(\alpha+\frac{5\beta}{2}\right)\ldots\cos\left(\alpha+n\beta-\frac{3\beta}{2}\right) - \cos\left(\alpha+(n-1)\beta+\frac{\beta}{2}\right)$$

$$2\sin\frac{\beta}{2}S = \cos\left(\alpha-\frac{\beta}{2}\right) - \cos\left(\alpha+(n-1)\beta+\frac{\beta}{2}\right)$$

$$\frac{2\sin\beta s}{2} = 2\sin\left(\frac{\alpha+(n-1)\beta+\frac{\beta}{2}-\alpha+\frac{\beta}{2}}{2}\right)\sin\left(\frac{\frac{\alpha-\beta}{12}+\alpha+(n-1)\beta+\frac{\beta}{2}}{2}\right)$$

$$\sin\frac{\beta}{2}s = \sin\left(\frac{n\beta}{2}\right)\sin\left(\frac{2\alpha+(n-1)\beta}{2}\right)$$

$$S = \frac{\sin\left(\frac{n\beta}{2}\right)\sin\left(\alpha+\frac{(n-1)\beta}{2}\right)}{\sin\frac{\beta}{2}}$$

(ii) $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots + \cos\{\alpha + (n-1)\beta\} = \dfrac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}} \cos\left(\alpha + \dfrac{n-1}{2}\beta\right)$

where : $\beta \neq 2m\pi$, $m \in I$

**Questions**

Simplify

1. $\sin\theta + \sin 2\theta + \sin 3\theta \ldots\ldots\ldots\ldots + \sin 19\theta.$

Ans. $\dfrac{\sin\left(\dfrac{19\theta}{2}\right)}{\sin\left(\dfrac{\theta}{2}\right)}\sin(10\theta)$

Sol. $\sin\theta + \sin 2\theta + \sin 3\theta \ldots\ldots\ldots + \sin 19\theta$

$\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) \ldots\ldots\ldots\ldots + \sin(\alpha + (n-1)\beta)$

$$= \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\sin\left(\alpha + \frac{(n-1)}{2}\beta\right)$$

$\alpha = \beta \quad \beta = \theta$

$$= \frac{\sin\frac{19\theta}{2}}{\sin\frac{\theta}{2}}\sin\left(\theta + \frac{18\theta}{2}\right)$$

$$= \frac{\sin\frac{19\theta}{2}}{\sin\frac{\theta}{2}}\sin\left(\frac{20\theta}{2}\right)$$

$$= \frac{\sin\frac{19\theta}{2}}{\sin\frac{\theta}{2}}\sin 10\theta$$

2. $\displaystyle\sum_{r=1}^{7} \cos\frac{r\pi}{7}$

Ans. $-1$

Sol. $\Rightarrow \cos\dfrac{\pi}{7} + \cos\dfrac{2\pi}{7} + \cos\dfrac{3\pi}{7} + \cdots + \cos\dfrac{7\pi}{7}$

$\alpha = \dfrac{\pi}{2} \quad \beta = \dfrac{\pi}{7} \quad \alpha + (x-1)\beta = \pi$

$$\frac{\pi}{7}+\frac{n\pi}{7}-\frac{x}{7}=x$$

$m=7$

$$\frac{\sin\frac{7\pi}{14}}{\sin\frac{\pi}{14}}\cos\left(\frac{\pi}{7}+\frac{7-1}{2}\frac{\pi}{7}\right) \qquad \Rightarrow \frac{\sin\frac{\pi}{2}}{\sin\frac{\pi}{14}}\cos\left(\frac{\pi}{7}+\frac{3\pi}{7}\right)$$

$$=\frac{1}{\sin\frac{\pi}{14}}\cos\left(\frac{4\pi}{7}\right) \qquad \Rightarrow \frac{\cos\left(\frac{\pi}{2}+\frac{\pi}{14}\right)}{\sin\frac{\pi}{14}}$$

$$=-\frac{\sin\frac{\pi}{14}}{\sin\frac{\pi}{14}}\Rightarrow -1$$

3. $\sum_{r=1}^{10} \sin^2(r\theta)$

Ans. $5-\frac{1}{2}\frac{\sin(10\theta)\cos 11\theta}{\sin\theta}$

Sol. $\sin^2\theta+\sin^2 2\theta+\sin^2 3\theta+\sin^2 4\theta$ .....$+\sin^2 10\theta$

$$\frac{1-\cos 2\theta}{2}+\frac{1-\cos 4\theta}{2}+\frac{1-\cos 6\theta}{2}+\ldots\ldots\ldots\ldots\frac{1-\cos 20\theta}{2}$$

$5-[\cos 2\theta+\cos 4\theta+\cos 6\theta$ ..... $\cos 20\theta]$

$\alpha=2\theta$ $\beta=2\theta$ $2\theta+(n-1)2\theta=20\theta$

$n=10$

$$5-\left[\frac{\sin 10\theta}{\sin\theta}\cos(2\theta+9\theta)\right] \qquad \Rightarrow 5-\frac{1}{2}\left[\frac{\sin 10\theta}{\sin\theta}\cos 11\theta\right]$$

4. $\sum_{r=1}^{10} \cos^2 r\theta$

Ans. $5+\frac{1}{2}\frac{\sin(10\theta)\cos 11\theta}{\sin\theta}$

Sol. $\cos^2\theta+\cos^2 2\theta+\cos^2 3\theta+$ .....$+\cos^2 10\theta$

$$\frac{1+\cos 2\theta}{2}+\frac{1+\cos 4\theta}{2}+\frac{1+\cos 6\theta}{2}+\cdots+\frac{1+\cos 20\theta}{2}$$

$$5+\frac{1}{2}\left(\frac{\sin 10\theta}{\sin\theta}\cos 11\theta\right)$$

5. $\sum_{r=1}^{10} \sin^3(r\theta)$

Ans. $\frac{3}{4}\left[\frac{\sin 5\theta \sin\left(\frac{11\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}\right] - \frac{1}{4}\left[\frac{\sin 15\theta}{\sin\left(\frac{3\theta}{2}\right)}\sin\left(\frac{33\theta}{2}\right)\right]$

Sol. $\sin^3\theta + \sin^3 2\theta + \sin^3 3\theta + \ldots\ldots\ldots + \sin^3 10\theta$

$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

$4\sin^3\theta = 3\sin\theta - \sin 3\theta$

$\sin^3\theta = \frac{3\sin\theta - \sin 3\theta}{4}$

$\frac{3\sin\theta - \sin 3\theta}{4} + \frac{3\sin 2\theta - \sin 6\theta}{4} + \frac{3\sin 3\theta - \sin 9\theta}{4} + \ldots\ldots + \frac{3\sin 10\theta - \sin 30\theta}{4}$

$\frac{1}{4}\left[3(\sin\theta + \sin 2\theta \ldots\ldots + \sin 10\theta) - (\sin 3\theta + \sin 6\theta + \sin 9\theta \ldots \sin 30\theta)\right]$

$\frac{3}{4}\left[\frac{\sin\frac{10\theta}{2}}{\sin\theta}\sin\left(\theta + \frac{9\theta}{2}\right)\right] - \frac{1}{4}\left[\frac{\sin\frac{30\theta}{2}}{\sin\frac{3\theta}{2}}\sin\left(3\theta + \frac{9(3\theta)}{2}\right]\right]$

$\frac{3}{4}\left[\frac{\sin 5\theta}{\sin\frac{\theta}{2}}\sin\left(\frac{11\theta}{2}\right)\right] - \frac{1}{4}\left[\frac{\sin 15\theta}{\sin 3\theta}\sin\left(\frac{33\theta}{2}\right)\right]\right]$

6. $\sin\theta - \sin 2\theta + \sin 3\theta - \sin 4\theta + \ldots\ldots\ldots\ldots\ldots$n terms.

Ans. $\frac{\sin\left(\frac{n\pi}{2} + \frac{\theta}{2}\right)}{\cos\frac{\theta}{2}}\sin\left(\frac{(n-1)\pi + (n+1)\theta}{2}\right)$

Sol. $\sin\theta + \sin(\pi + 2\theta) + \sin(2\pi + 3\theta) + \sin(3\pi + 4\theta) + \ldots\ldots\ldots + \sin[(n-1)\pi + n\theta]$

$\beta = c.\alpha = \pi + \theta \qquad \alpha = \theta$

$\frac{\sin\frac{n\pi + \theta}{2}}{\sin\frac{\pi + \theta}{2}}\sin\left\{\theta + \frac{n-1}{2}(\pi + \theta)\right\}$

$$\frac{\sin\left(\frac{n\pi}{2}+\frac{\theta}{2}\right)}{\cos\frac{\theta}{2}}\sin\left\{\frac{(n-1)\pi}{2}+\left(1+\frac{n-1}{2}\right)\theta\right\}$$

$$\frac{\sin\left(\frac{n\pi}{2}+\frac{\theta}{2}\right)}{\cos\frac{\theta}{2}}\sin\left\{\left(\frac{n-1}{2}\right)\pi+\left(\frac{n+1}{2}\right)\theta\right\}$$

## 8.2 SUMMATION OF SERIES TYPE - 2

$\frac{\sin(A-B)}{\cos A\cos B}=\tan A-\tan B$ ; $\frac{\sin(A-B)}{\sin A\sin B}=\cot B-\cot A$

$T_n=f(n+1)-f(n)$

$T_1=f(2)-f(1)$

$T_2=f(3)-f(2)$

$T_3=f(4)-f(3)$

| | | |

| | | |

$T_n=f(n+1)-f(n)$

sum = $f(n+1)-f(1)$

Q.1 $\sum_{r=0}^{44}\frac{1}{\cos(r^\circ).\cos(r+1)^\circ}$

Ans $\frac{1}{\sin 1^\circ}$

Sol. $T_1=\frac{1}{\cos 0^\circ\cos 1^\circ}$

$=\frac{1}{\sin 1^\circ}\frac{\sin(1^\circ-0^\circ)}{\cos 0^\circ\cos 1^\circ}$

$=\frac{1}{\sin 1^\circ}(\tan 1^\circ-\tan 0^\circ)$

$T_2=\frac{1}{\sin 1^\circ}(\tan 2^\circ-\tan 1^\circ)$

$T_{45}=\frac{1}{\sin 1^\circ}(\tan 45^\circ-\tan 44^\circ)$

$$\Rightarrow \quad \sum_{r=0}^{44} \frac{1}{\cos(r^\circ).\cos(r+1)^\circ}$$

$$= T_1 + .....+ T_{45}$$

$$= \frac{1}{\sin 1^\circ}\left(\tan 45^\circ\right) = \frac{1}{\sin 1^\circ}$$

Q.2 $\sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right).$

Ans. $\tan x - \tan\left(\frac{x}{2^n}\right)$

Sol. $T_1 = \tan\frac{x}{2}\sec x$

$$\frac{\sin\left(x-\frac{x}{2}\right)}{\cos\frac{x}{2}\cos x} = \left(\tan x - \tan\frac{x}{2}\right)$$

$$T_2 = \tan\frac{x}{4}\sec\frac{x}{2}$$

$$\frac{\sin\left(\frac{x}{2}-\frac{x}{4}\right)}{\cos\frac{x}{2}\cos\frac{x}{4}} = \left(\tan\frac{x}{2} - \tan\frac{x}{4}\right)$$

$$T_3 = \tan\frac{x}{4} - \tan\frac{x}{8}$$

$$\vdots$$

$$T_n = \tan\frac{x}{2^{n-1}} - \tan\frac{x}{2^n}$$

$$\sum T_m = \tan x - \tan\frac{x}{2^n}$$

Q.3 $\sum_{r=1}^{n} \operatorname{cosec}\left(2^{r-1}x\right)$ .

Ans. $\cot\left(\frac{x}{2}\right) - \cot\left(2^{n-1}x\right)$

Sol. $T_1 = \operatorname{cosec} x \rightarrow \frac{1}{\sin x}$ $\qquad$ $T_2 = \operatorname{cosec} 2x = \frac{1}{\sin 2x}$

$$= \frac{\sin\left(x - \frac{x}{2}\right)}{\sin x \sin\frac{x}{2}} \qquad = \frac{\sin(2x - x)}{\sin 2x \sin x}$$

$$= \cot\frac{x}{2} - \cot x \qquad = \cot x - \cot 2x$$

$T_3 = \text{cosec } 4x$

$$= \frac{1}{\sin 4x} \Rightarrow \frac{\sin(4x - 2x)}{\sin 2x \sin 4x}$$

$= \cot 2x - \cot 4x$

$\vdots$

$T_n = \cot 2^{n-2}x - \cot 2^{n-1}x$

$$\sum T_n = \cot\frac{x}{2} - \cot 2^{n-1}x$$

## 9.1 INEQUALITIES IN A TRIANGLE

1. Prove that $\cot^2 A + \cot^2 B + \cot^2 C \geq 1$.

Sol. $A + B + C = \pi$

$\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

Let $\cot A = x \quad \cot B = y \quad \cot C = z$

$xy + yz + zx = 1$

$(x - y)^2 + (y - z)^2 + (y - z)^2 \geq 0$

$2x^2 + 2y^2 + 2z^2 - 2(xy + yz + zx) \geq 0$

$x^2 + y^2 + z^2 \geq 1$

$\cot^2 A + \cot^2 B + \cot^2 C \geq 1$

2. Prove that $\cos A + \cos B + \cos C \leq \frac{3}{2}$.

Sol. $A + B + C = \pi$

$$\frac{A + B}{2} = \frac{\pi}{2} - \frac{C}{2}$$

$\cos A + \cos B + \cos c = x$

$$2\cos\frac{A + B}{2}\cos\left(\frac{A - B}{2}\right) + 1 - 2\sin^2\frac{C}{2} = x$$

$$2\cos\left(\frac{\pi}{2} - \frac{C}{2}\right)\cos\left(\frac{A - B}{2}\right) - 2\sin^2\frac{C}{2} + 1 = x$$

$$2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right)-2\sin^2\frac{C}{2}+1=x$$

$$2\sin^2\frac{C}{2}-2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right)+(x-1)=0$$

$D\geq 0$

$$4\cos^2\left(\frac{A-B}{2}\right)-4(2)(x-1)\geq 0$$

$$\cos^2\left(\frac{A-B}{2}\right)\geq 2(x-1)$$

$$2(x-1)\leq\cos^2\left(\frac{A-B}{2}\right)\leq 1$$

$2(x-1)\leq 1$

$2x\leq 3$

$$x\leq\frac{3}{2}$$

$$\cos A+\cos B+\cos C\leq\frac{3}{2}$$

## 9.2 TRIGONOMETRIC SUBSTITUTION

1. If $x^2+y^2=r^2$ put $x=r\cos\theta,\ y=r\sin\theta$

Q.1 $x^2+y^2=9$ find range of $3x+4y$

Ans $[-15,15]$

Sol $x=3\cos\theta$

$y=3\sin\theta$

$3x+4y=9\cos\theta+12\sin\theta$

$=[-15,15]$

2. (i) $\sqrt{a^2-x^2}$ $x=a\sin\theta,\ x=a\cos\theta$ (ii) $\sqrt{a^2+x^2}$ $x=a\tan\theta$

(iii) $\sqrt{x^2-a^2}$ $x=a\sec\theta$

(iv) $\frac{2x}{1-x^2}$ $x=\tan\theta$ (v) $\frac{1-x^2}{1+x^2}$ $x=\tan\theta$

(vi) $\sin^{-1}(4x^3-3x)$ $x=\cos\theta$ (vii) $\frac{1-x}{1+x}$ $x=\cos\theta$